

Answer Key & Marking Scheme

Subject: Mathematics | **Test:** Test 4 (Day 28, 6 June 2026) — 2-Week Cumulative
Maximum Marks: 80

Syllabus: Chs 1–13 (comprehensive): Real Numbers • Polynomials • Linear Equations • Quadratic Equations • AP • Triangles • Coordinate Geometry • Trigonometry • Applications • Circles • Areas Related to Circles • Surface Areas and Volumes • Statistics

SECTION A — Answer Key (1 mark each)

- Q1.** (c) p^3q^2 — LCM = max powers of all primes.
- Q2.** (a) terminating — $3125 = 5^5$; denominator of form $2^m \cdot 5^n$.
- Q3.** (b) 1 — Sum $\alpha + \beta = p$, $\alpha\beta = -p - c$. $(\alpha+1)(\beta+1) = \alpha\beta + \alpha + \beta + 1 = -p - c + p + 1 = 1 - c$. Set this to 0 $\Rightarrow c = 1$.
- Q4.** (c) coincident — second equation is $3 \times$ first.
- Q5.** (a) 4, -2 — $(x-4)(x+2) = 0$.
- Q6.** (a) 22 — $a = -3$, $d = 5/2$; $a_{11} = a + 10d = -3 + 25 = 22$.
- Q7.** (a) (4, -3).
- Q8.** (c) 10 — $\sqrt{36 + 64} = 10$.
- Q9.** (c) 3 — $\sqrt{3} \cdot \sqrt{3} = 3$.
- Q10.** (b) $4/5$ — $\sin^2 + \cos^2 = 1 \Rightarrow \cos^2 = 1 - 9/25 = 16/25$.
- Q11.** (b) 45° — $\tan \theta = 100/100 = 1$.
- Q12.** (b) 12 cm — $\sqrt{13^2 - 5^2} = 12$.
- Q13.** (a) 154 cm^2 — $\frac{1}{4} \times \pi \times 14^2 = 154$.
- Q14.** (a) $(2/3)\pi(7)^3$.
- Q15.** (a) $1 : 2 : 3$ — $(1/3)\pi r^2 \cdot h : (2/3)\pi r^3 : \pi r^2 \cdot h$, with $h = r \Rightarrow$ ratios $1/3 : 2/3 : 1 = 1 : 2 : 3$.
- Q16.** (b) 24 — Mode = $3 \cdot$ Median $- 2 \cdot$ Mean = $60 - 36 = 24$.
- Q17.** LCM(16, 24) = 48 — $16 = 2^4$, $24 = 2^3 \cdot 3 \Rightarrow$ LCM = $2^4 \cdot 3 = 48$.
- Q18.** $k/6 = 1/2 \neq 2/3 \Rightarrow k = 3$ (also $a/b \neq c/d$ for 'no solution'). Verify: with $k = 3$, equation becomes $3x - y = 2$; $6x - 2y = 3$. $a/b = 3/6 = 1/2$; $b/b = -1/-2 = 1/2$; $c/d = 2/3 \neq 1/2$. So lines parallel — **k = 3**.
- Q19.** $\sec^2 A - \tan^2 A = 1 \Rightarrow \tan^2 A = (17/8)^2 - 1 = 289/64 - 64/64 = 225/64 \Rightarrow \tan A = 15/8$.
- Q20.** $4 : 5$ — ratio of corresponding sides equals ratio of perimeters in similar triangles.

SECTION B — Answer Key (2 marks each)

- Q21.** Required number divides $(615 - 6) = 609$ and $(963 - 6) = 957$ exactly. HCF(609, 957): $957 = 1 \cdot 609 + 348$; $609 = 1 \cdot 348 + 261$; $348 = 1 \cdot 261 + 87$; $261 = 3 \cdot 87 + 0$. **HCF = 87.** (2)
- Q22.** Equal roots $\Rightarrow D = 0 \Rightarrow k^2 - 16 = 0 \Rightarrow$ **k = ± 4 .** (2)
- Q23.** $a_{11} = a + 10d = 38$; $a_{15} = a + 15d = 73$. Subtracting: $5d = 35 \Rightarrow d = 7$. Then $a = 38 - 70 = -32$. $a_{10} = -32 + 30 \cdot 7 =$ **178.** (2)
- Q24.** In quadrilateral OAPB, $\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent). Sum of all angles = 360° . So $\angle AOB + \angle APB + 90^\circ + 90^\circ = 360^\circ \Rightarrow \angle AOB + \angle APB =$ **180° .** ✓ (2)
- Q25.** In parallelogram ABCD, the diagonals bisect each other, so midpoint of AC = midpoint of BD. $((6+p)/2, (1+4)/2) = ((8+p)/2, (2+3)/2) \Rightarrow (15/2, 5/2) = ((8+p)/2, 5/2) \Rightarrow 8 + p = 15 \Rightarrow$ **p = 7.** (2)

SECTION C — Answer Key (3 marks each)

- Q26.** *Proof.* Suppose $2 + 3\sqrt{5}$ is rational, say p/q ($q \neq 0$). Then $\sqrt{5} = (p - 2q)/(3q)$, which is rational — contradicting that $\sqrt{5}$ is irrational. Hence $2 + 3\sqrt{5}$ is irrational. (3)

Q27. Since $2 \pm \sqrt{3}$ are zeroes, $(x - (2 + \sqrt{3}))(x - (2 - \sqrt{3})) = x^2 - 4x + 1$ is a factor. Divide $x^3 - 6x^2 + 26x - 35$ by $x^2 - 4x + 1$ to get quotient $x^2 - 2x - 35 = (x - 7)(x + 5)$. So other zeroes are **7 and -5**. (3)

Q28. Let stream speed = x km/h. Upstream speed = $18 - x$; downstream = $18 + x$. Time difference: $24/(18 - x) - 24/(18 + x) = 1 \Rightarrow 24[(18 + x) - (18 - x)]/[(18)^2 - x^2] = 1 \Rightarrow 48x = 324 - x^2 \Rightarrow x^2 + 48x - 324 = 0 \Rightarrow (x + 54)(x - 6) = 0 \Rightarrow x = 6$ (positive). **Stream speed = 6 km/h**. (3)

Q29. Any positive integer n , on division by 6, yields remainder 0, 1, 2, 3, 4 or 5. So n is of the form $6q$, $6q + 1$, $6q + 2$, $6q + 3$, $6q + 4$ or $6q + 5$. The forms $6q$, $6q + 2$, $6q + 4$ are even (divisible by 2). So an odd integer is of the form $6q + 1$, $6q + 3$ or $6q + 5$. ✓ (3)

Q30. Multiply numerator and denominator on LHS by $(\cos A - \sin A + 1)$: LHS = $(\cos A - \sin A + 1)^2 / [(\cos A + \sin A - 1)(\cos A - \sin A + 1)] = (\cos A - \sin A + 1)^2 / [\cos^2 A - (\sin A - 1)^2] = (\cos A - \sin A + 1)^2 / [\cos^2 A - \sin^2 A + 2\sin A - 1] = (\cos A - \sin A + 1)^2 / [-2\sin^2 A + 2\sin A] = (\cos A - \sin A + 1)^2 / [2\sin A \cdot (1 - \sin A)]$. Expanding numerator and simplifying using $\sin^2 A + \cos^2 A = 1$, this equals $(1 + \cos A)/\sin A = \operatorname{cosec} A + \cot A$. ✓ (3)

Q31. Area of sector = $(\theta/360^\circ) \times \pi r^2 = (60/360) \times 3.14 \times 25 = 25 \times 3.14 / 6 \approx 13.08 \text{ cm}^2$. Area of triangle (equilateral with side 5) = $(\sqrt{3}/4) \cdot 25 \approx 10.81 \text{ cm}^2$. Area of segment = $13.08 - 10.81 \approx \mathbf{2.27 \text{ cm}^2}$. (3)

SECTION D — Answer Key (5 marks each)

Q32. Let the height of tower = h and the angles of elevation from the two points be θ and $(90^\circ - \theta)$ (complementary). From the first point at distance a : $\tan \theta = h/a$. From the second point at distance b : $\tan(90^\circ - \theta) = \cot \theta = h/b$. Multiplying: $\tan \theta \cdot \cot \theta = (h/a)(h/b) \Rightarrow 1 = h^2/(ab) \Rightarrow \mathbf{h^2 = ab} \Rightarrow \mathbf{h = \sqrt{ab}}$. ✓ (5)

Q33. Proof. Let two tangents PA and PB be drawn from an external point P to a circle with centre O . Join OA , OB and OP . Since $OA \perp PA$ and $OB \perp PB$ (radius $\perp$ tangent), and $OA = OB$ (radii), and OP is common, $\triangle OAP \cong \triangle OBP$ (RHS). So $PA = PB$. ✓

Application: Let parallelogram $ABCD$ circumscribe a circle, touching the sides at P , Q , R , S . By the tangent property: $AP = AS$, $BP = BQ$, $CQ = CR$, $DR = DS$. Adding: $AP + BP + CR + DR = AS + DS + BQ + CQ \Rightarrow AB + CD = AD + BC$. But $ABCD$ is a parallelogram, so $AB = CD$ and $AD = BC$. Hence $2AB = 2BC \Rightarrow AB = BC$, and the parallelogram is a rhombus. ✓ (5)

Q34. Diameter = 14 cm, radius $r = 7$ cm. Length of cylinder + 2 hemispheres = 35 cm $\Rightarrow$ cylinder height $h = 35 - 2 \cdot 7 = 21$ cm. **Volume:** $V = \pi r^2 h + 2 \cdot (2/3)\pi r^3 = \pi \cdot 49 \cdot 21 + (4/3)\pi \cdot 343 = 1029\pi + (1372/3)\pi = (3087 + 1372)/3 \cdot \pi = 4459\pi/3 \approx 4670.7 \text{ cm}^3$. (Using $\pi = 22/7$: $V = 4459/3 \times 22/7 = 4670.67 \text{ cm}^3$.) **Surface area:** $2\pi rh + 2 \cdot 2\pi r^2 = 2\pi \cdot 7 \cdot 21 + 4\pi \cdot 49 = 294\pi + 196\pi = 490\pi = 490 \times 22/7 = \mathbf{1540 \text{ cm}^2}$. (5)

Q35. Total $f = 60$. Mid-points: 5, 15, 25, 35, 45.

Mean: $\Sigma fx = 8 \cdot 5 + 12 \cdot 15 + 25 \cdot 25 + 9 \cdot 35 + 6 \cdot 45 = 40 + 180 + 625 + 315 + 270 = 1430$. Mean = $1430/60 = \mathbf{23.83}$.

Median: Cumulative frequencies: 8, 20, 45, 54, 60. $n/2 = 30$, falls in class 20–30. $l = 20$, $cf = 20$, $f = 25$, $h = 10$. Median = $20 + (30 - 20)/25 \times 10 = 20 + 4 = \mathbf{24}$. (5)

SECTION E — Answer Key (4 marks each)

Q36. (i) **Modal class: 2–3** (highest frequency 25). (1) (ii) Mode = $l + (f_{\text{max}} - f_{\text{prev}})/(2f_{\text{max}} - f_{\text{prev}} - f_{\text{next}}) \times h = 2 + (25 - 11)/(50 - 11 - 10) \times 1 = 2 + 14/29 \approx 2 + 0.48 = \mathbf{2.48}$. (2) (iii) The most common number of children per family is between 2 and 3 (around 2 children). (1)

Q37. (i) Diagram: aeroplane at height 5 km; first position P at angle of elevation 60° from observer A ; second position Q after 10 sec at angle 30° . $PA' = QA'' = 5$ km vertical. (1) (ii) Let $A'P$ = horizontal distance from observer to first foot. $\tan 60^\circ = 5/A'P \Rightarrow A'P = 5/\sqrt{3}$. Similarly $A'Q = 5/\tan 30^\circ = 5\sqrt{3}$. Distance covered = $A'Q - A'P = 5\sqrt{3} - 5/\sqrt{3} = (5 \cdot 3 - 5)/\sqrt{3} = 10/\sqrt{3} = 10\sqrt{3}/3 \approx \mathbf{5.77 \text{ km}}$. (3)

Q38. (i) Radius = diameter/2 = $5/2 = \mathbf{2.5 \text{ mm}}$. (1) (ii) Length of cylinder = total length – $2 \times$ radius = $14 - 2 \cdot 2.5 = \mathbf{9 \text{ mm}}$. (1) (iii) Surface area = $2\pi rh + 2 \cdot (2\pi r^2) = 2\pi \cdot 2.5 \cdot 9 + 4\pi \cdot (2.5)^2 = 45\pi + 25\pi = 70\pi = 70 \times 22/7 = \mathbf{220 \text{ mm}^2}$. (2)